
EE-5102/CS-6304 — Advanced Topics in Machine Learning

Programming Assignment 1

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Abstract

In general, supervised learning solutions generally assume that training and test data are independent and identically distributed. These solutions also assume a closed-set setting: a classification problem where the model only encounters classes during testing that it was explicitly trained to recognise. However, this is normally not the case. An object from a known class may appear with altered textures, colors, positions, or styles; test data may come from new domains; or an input may belong to a class that was not observed during training. Consequently, a model can perform well in a controlled environment but break down in the real world because it relies on cues or statistical regularities that do not remain stable. This report examines aspects of learning beyond the aforementioned assumptions, and then forms an argument about what a useful representation should preserve, suppress, and reject. You may find the associated code and $\text{\LaTeX}$ files for this assignment in the following GitHub repository: <https://github.com/FaaizKhan1999/atml-assignment1>

1 Tasks and Experiments

1.1 Inductive Biases and Feature Representations

This is something I need to cite for this subsection [1].

1.2 Unsupervised Domain Adaptation

1.3 Domain Generalisation

1.4 Open-Set Recognition (OSR)

Table 1: Baseline ResNet-152 training and validation performance

Epoch	Performance	
	Training Loss	Validation Accuracy
1	0.6616	81.73%
2	0.5318	83.62%
3	0.4901	83.63%
4	0.4721	83.77%
5	0.4653	84.38%

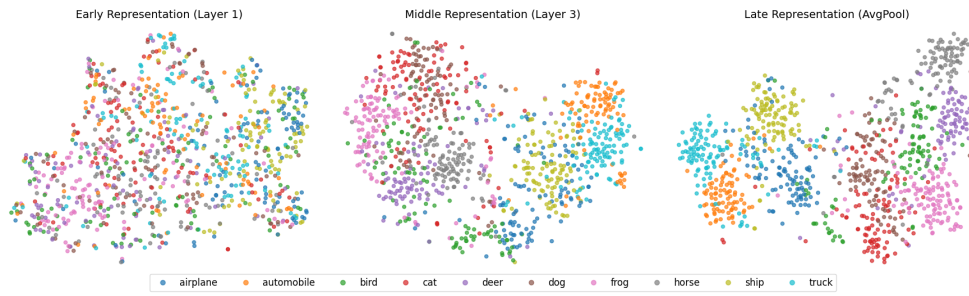


Figure 1: Feature visualisation of early, middle, and late layers in a ResNet-152 model evaluated on CIFAR-10

References

- [1] Robert Geirhos, Patricia Rubisch, Claudio Michaelis, Matthias Bethge, Felix A. Wichmann, and Wieland Brendel. Imagenet-trained cnns are biased towards texture; increasing shape bias improves accuracy and robustness, 2022. URL <https://arxiv.org/abs/1811.12231>.